Solids

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Mechanics of Solids

Indifferent from rigid body dynamics, in mechanics of solids we study the effect of forces on the bodies causing deformation in them. So, all the bodies are non-rigid bodies in mechanics of solids.
Mechanics of solids is study of behavior of materials under compressive (squashing) and tensile (stretching) forces.
In Engineering, it is essential to find the behavior of materials to know the suitability of a particular material for a particular job. E.g., In bridges, the beams, rivets and other members may be subjected to tension and compression. In order for the bridge to last long, these stresses must be accounted for.
Some terms needed to be known before studying mechanics of solids are given below-

DEFORMING FORCE

In rigid bodies, the external forces can produce translational acceleration or angular acceleration. It is so because the rigid bodies are considered to be non-deformable.
But no physical body is completely rigid. So, forces can also deform bodies.
If a force applied on a body produces a change in the normal positions of the molecules of the body, it is called deforming force.

ELASTICITY

The property of a body which makes it to regain its original configuration when the deforming forces acting on it are removed is known as elasticity.

PERFECTLY ELASTIC BODY

After the removal of deforming forces, if a body completely gains its original configuration, then it is known as a perfectly elastic body.

PERFECTLY PLASTIC BODY

Even on applying a very small deforming force, if the body is not able to regain its original configuration after the removal of the deforming force, then it is known as perfectly plastic body.

RESTORING FORCE

Restoring force is the opposing force to the applied deforming force. Restoring force arises due to elasticity of the body and it tries to restore the original shape of the body.

STRESS AND STRAIN

It is imperative to study the behavior of materials under tension, compression, bending, torsion etc as it describes at what loading conditions, a material can break. In machines and structures, the failure of a component can lead to wastage of production or lives. So, the study of deformation of bodies under loading becomes necessary.

Figure shown below illustrates four examples, showing tensile force, shearing force and volumetric compression.
For each kind of deformation, a quantity called stress that characterizes the strength of the force per unit area and another quantity, strain, which describes the resulting deformation will be defined.

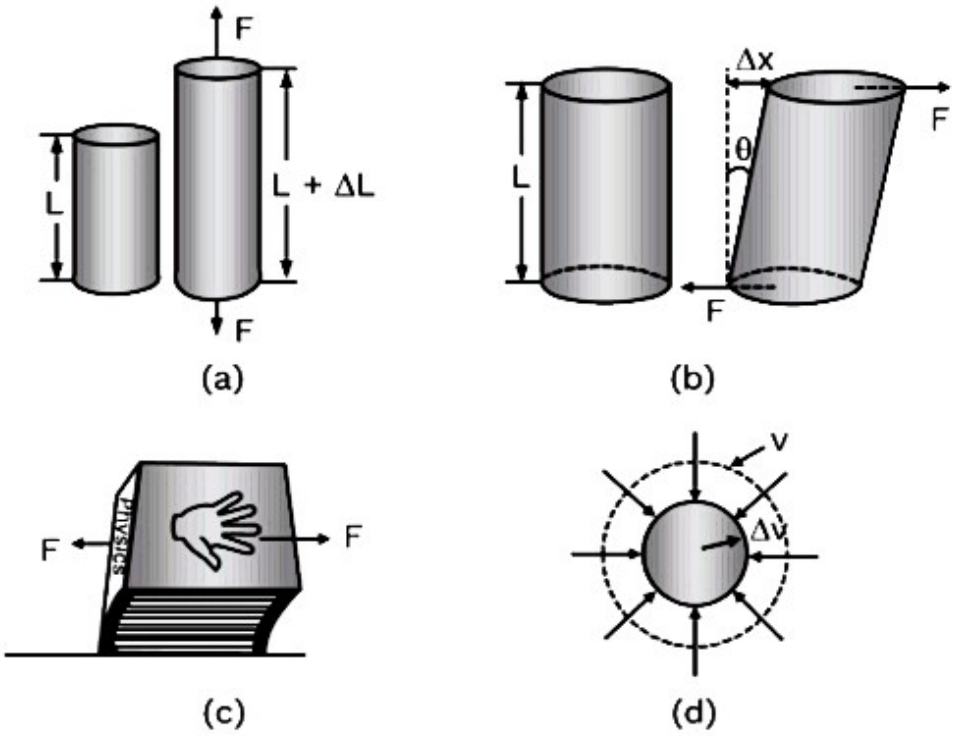


Figure: (a) A rod subjected to tensile stress stretches by length ΔL.
(b) A cylinder subjected to shearing (tangential) stress deforms it by an angle θ.
(c) A book subjected to a shearing stress
(d) A solid sphere subjected to a uniform hydraulic stress shrinks in volume by an amount ΔV.

STRAIN

Strain is the ratio of change in a dimension to the original dimension of the body due to application of external forces.

Strain is a unit less quantity.

If an external force is applied on a deformable body, it tries to produce deformation in it. The forces internal to the body try to oppose this deformation causing a stress in the body. The opposing or restoring force produced inside the body is equal and opposite to the externally applied deforming force. So, the restoring force is the stress causing force in the body.

STRESS

The restoring force (opposing force to deforming force) developed in the body per unit area is termed as stress.

Stress = Restoring Force/ Area

At equilibrium, restoring force is equal to external deforming force, so we have

Stress = External deforming force / Area

Unit: In SI system, stress is measured in N/m^2 and in CGS system, it is measured in $dyne/cm^2$

The dimensional formula for stress is **$[M L^{-1} T^{-2}]$**

TENSILE AND COMPRESSIVE STRESS AND STRAIN

The most common and easiest example to study elastic behavior is stretching of a rod or wire by pulling its ends.

Figure shows an object having a uniform area of cross-section A initially and length l_0. Now, forces of equal magnitude $F_\perp$ but in opposite directions are applied to the object as shown in figure below at its ends. As equal and opposite forces act on its ends, so the object cannot move in any direction.

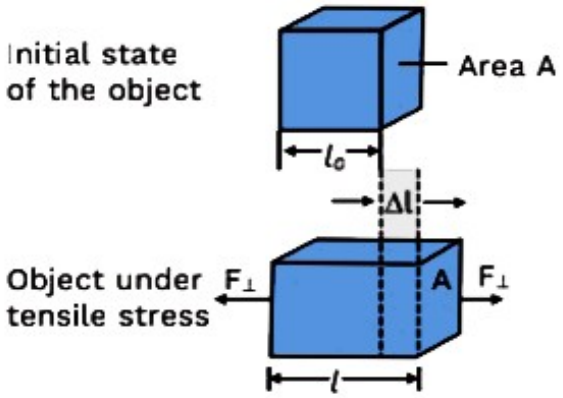


Fig. : *An object in tension. The net force on the object is zero, but the object deforms. The tensile stress (the ratio of the force to the cross-sectional area) produces a tensile strain (the elongation divided by the initial length). The elongation Δl is exaggerated for clarity.*

Tensile stress on the object is defined as,

$$\text{Tensile stress} = \frac{F_\perp}{A}$$

Stress is a scalar quantity and we use the magnitude of the force ($F_\perp$) only.

"pascal" is SI unit of stress.

$$\boxed{1 \text{ pascal} = 1\text{Pa} = 1\ \text{N}/\text{m}^2}$$

As a result of the tensile force acting on the given bar, it stretches to a length $l = l_0 + \Delta l$. The elongation Δl is uniform for every part of the bar.

$$\text{Tensile strain} = \frac{\Delta l}{l_0}$$

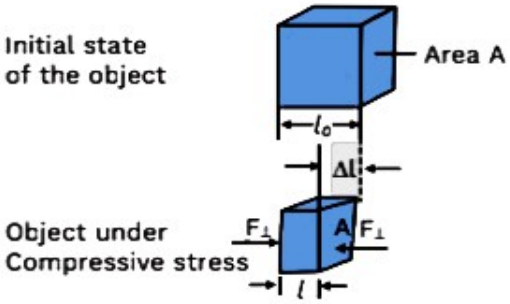


Fig. *: An object in compression. The compressive stress and compressive strain are defined in the same way as tensile stress and strain, except that Δl now denotes the length through which the object contracts.*

When the forces at the ends of a bar are compressing it, the stress is compressive stress. The compressive stress is defined similarly as the tensile stress, but Δl has the opposite direction.

Q. **A rod has a radius of 50 mm and a length of 100 cm. A 1000N force is compressing it along its length. Find the stress developed in the rod**

Sol. Given, $r = 50\ mm = 50 \times 10^{-3}\ m$

$F = 1000\ N$

$$\text{Stress} = \frac{\text{Force}}{\text{Area}}$$

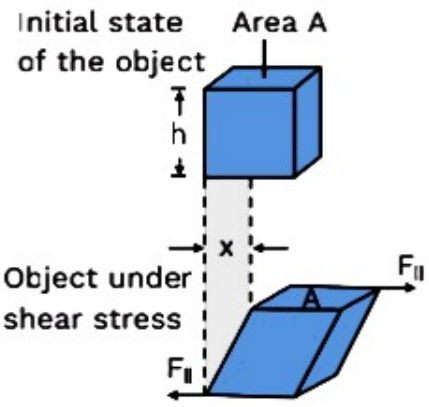
$$= \frac{1000N}{\pi(50 \times 10^{-3})^2 m^2} = 1.27 \times 10^5\ N/m^2$$

SHEAR STRESS AND STRAIN

The adjacent figure shows a body acted upon by forces of equal magnitude but opposite directions tangential to the surface of opposite ends of the object.

Shear stress is defined as the ratio of tangential force $F_\parallel$ and the area A on which it acts.

$$\text{Shear stress} = \frac{F_\parallel}{A}$$



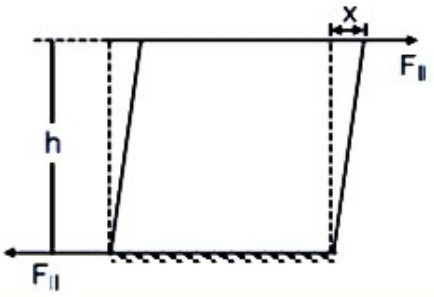
Due to application of tangential force, one face of the object under shear stress is displaced by a distance x relative to the opposite face.

Shear strain is defined as ratio of the displacement x to the transverse length h.

Shear strain = $\frac{x}{h}$

In real-life situation, x is nearly always much smaller than h. Like all strains, shear strain is a dimensionless number, it is a ratio of two lengths.

Q. **Consider a brass plate of height 0.80 m and 0.50 m width fixed from its lower end. A shear force acts on it due to an earthquake. What is the shear strain in the plate if the displacement x in upper face of plate is 0.16 mm.**



Sol. Shear strain $= \frac{x}{h}$

$= \frac{0.16 \times 10^{-3}\,m}{0.80m}$

$= 2.0 \times 10^{-4}$

BULK STRESS AND STRAIN

When a deep sea diver dives deep into ocean, the ocean water exerts nearly uniform pressure everywhere on his body surface and compresses him to a slightly smaller volume.
The stress acting in such condition is a uniform pressure on whole surface and the deformation occurs in form of change in the volume.

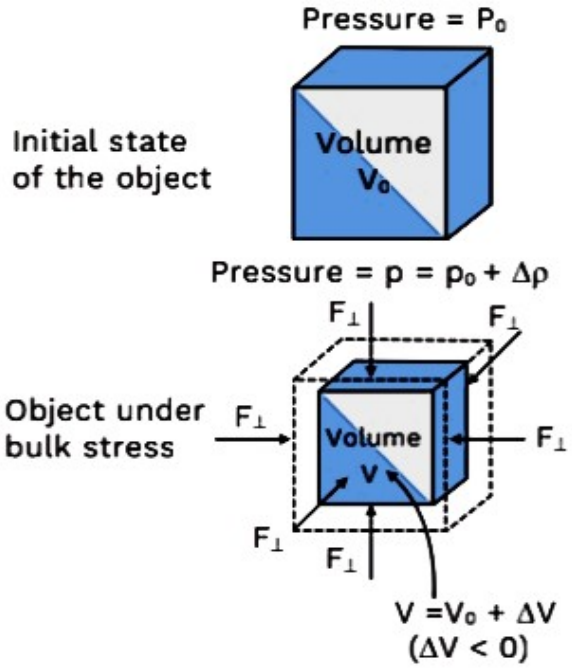


Fig. *: An object under bulk stress. Without the stress, the cube has volume V_0. When the stress is applied, the cube has a smaller volume V. The volume change ΔV is exaggerated for clarity.*

A fluid (liquid or gas) at rest can exert a perpendicular force on the immersed surface. The perpendicular force per unit area on the immersed surface is called pressure of the fluid.

$$p = \frac{F_\perp}{A} \text{ (Pressure in a fluid)}$$

The pressure in a fluid increases with increase in depth.

Pressure has the same unit as that of stress. Unit of pressure is 1 Pa = 1 N/m^2.

One more unit of pressure is atmosphere (abbreviated as atm). 1 atm is the approximate average pressure of the earth's atmosphere at sea level.

1 atmosphere = 1 atm = 1.013×10^5 Pa

Pressure acts as stress during volumetric deformation.
Volumetric strain is given by the ratio of change in volume (ΔV) to the original volume V_0.

$$\text{Bulk (volume) strain} = \frac{\Delta V}{V_0}$$

Q. **A spherical ball contracts in volume by 0.05%, when subjected to a normal uniform pressure. Calculate the volume strain produced in spherical ball.**

Sol.

$$\text{Volume strain} = \frac{\Delta V}{V}$$

$$= \frac{0.05}{100}$$

$$= 5 \times 10^{-4}$$

HOOKE'S LAW

For small enough values of stress and strain, stress is directly proportional to strain.

The proportionality constant is called **elastic modulus**.

Greater the pull applied on a specimen, the more it stretches.

The more it is squeezed, the more it compresses.

This relationship can be expressed as,

$$\frac{\text{Stress}}{\text{Strain}} = \text{Elastic modulus (Hooke's Law)}$$

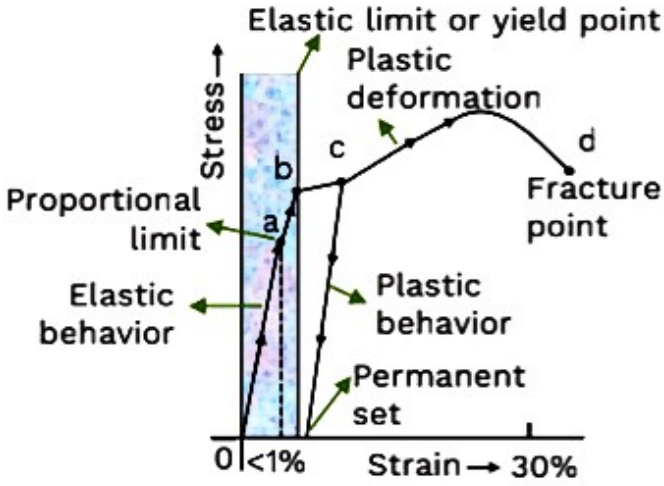
The proportionality of stress and strain (under certain condition) is called Hooke's law, after Robert Hooke (1635-1703).

Hooke's law is not actually a general law but an experimental finding that is valid over only a limited range.

STRESS-STRAIN CURVE

As per Hooke's law, stress is directly proportional to strain within the elastic limit of the specimen being tested.

For a very limited range, proportionality of stress and strain is valid in elastic deformations.



A stress-strain diagram for a specimen is plotted by fitting it in a Universal Testing Machine (UTM). The stress on the specimen is varied by the machine and the resulting strain is noted. With help of the values, the graph is plotted.

When Hooke's law is obeyed, the stress-strain graph is a straight line with the slope equal to an elastic modulus.

Figure shows a typical stress-strain graph for a ductile metal specimen such as copper or soft iron. The strain is shown as the percentage elongation on the x-axis.

From the graph, we can see that for a strain of less than 1%, the stress is directly proportional to strain. The straight-line variation ends at point a in the graph and the value of stress at point "a" is termed as the proportional limit of the specimen. If the specimen is unloaded in the region o-a, it regains its original configuration.

From a to b region in the stress-strain graph, strain is no longer proportional to stress and Hooke's law is not obeyed. But if the load is gradually removed, starting between o and b, the specimen regains its original configuration and the curve can be retraced back. From o-a-b, the deformation produced is reversible, and the forces are conservative in nature. The energy given to the material to produce the deformation is recovered after the removal of the stress. In region o-b, the material shows the elastic behavior but the Hooke's law is not obeyed in region a-b. Point b is known as yield point. The stress at the yield point is called elastic limit of the specimen.

On increasing stress beyond point b, the strain continues to increase. But now when we remove the load at some point beyond b, say c, then the material does not regain its original length. Instead, it follows the plastic behavior line in figure. The length at zero stress is now greater than the original length. The material has undergone an irreversible deformation and has acquired what we call a permanent set.

Further, increase of load beyond c produces a large increase in strain for a relatively small increase in stress, until a point d is reached at which fracture takes place. The behavior of the material from b to d is called plastic deformation. A plastic deformation is an irreversible deformation. After removing the stress, the specimen cannot return to its original state.

A large plastic deformation occurs in between elastic limit and fracture point. This property of having a large gap between elastic point and the fracture point makes the material to be a ductile one. If there is very small gap on stress-strain curve between the elastic limit and the fracture point, then the material is said to be brittle. A soft iron wire that can have considerable permanent stretch without breaking is ductile, while a steel piano string that breaks soon after its elastic limit is reached is brittle.

The stress needed to cause actual fracture of material is called the ultimate stress or breaking stress.

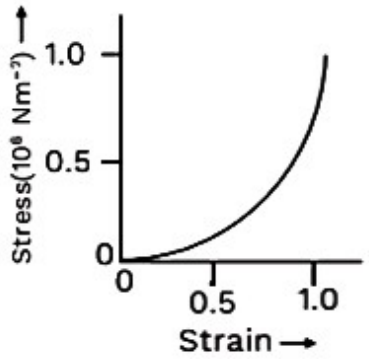
Table : Approximate Breaking Stresses

Material	Breaking Stress (Pa or N/m^2)
Steel	$5\text{-}20\times10^8$
Phosphor bronze	5.6×10^8
Glass	10×10^8
Iron	3.0×10^8
Brass	4.7×10^8
Aluminium	2.2×10^8

The stress-strain curve is different for different kinds of material. e.g.; A rubber band can be many times of its actual length and still it regains its original shape.

Materials like rubber or aorta tissues do not obey Hooke's law and they have no well defined plastic region.

Substances which can be elongated to large strains are called **elastomers.** Stress-strain curve for an elastomer is shown below.



ELASTIC AFTER EFFECT

After removing the deforming force, the time lag taken by the body to regain its original shape is termed as elastic after effect.
It is negligibly small for quartz and phosphor bronze, but very large for glass fibre.

ELASTIC FATIGUE

The loss of elastic nature of a body under application of repeated alternating deforming forces is termed as elastic fatigue.

Breaking force = Breaking stress × area of cross section of the wire.

Breaking stress is fixed for a material, but breaking force will vary, depending on area of cross section of the wire.

ELASTIC MODULI

The ratio of stress and strain is called modulus of elasticity, and it is a characteristic of the material.

Young's Modulus

For a sufficiently small tensile stress, stress and strain are directly proportional.
The corresponding elastic modulus is known as Young's modulus, denoted by Y :

$$Y = \frac{\text{Tensile stress}}{\text{Tensile strain}} = \frac{\left(\frac{F}{A}\right)}{\left(\frac{\Delta l}{l_0}\right)} = \left(\frac{F \cdot l_0}{A \cdot \Delta l}\right)$$

Unit of Young's modulus is same as that of stress.
Some typical values of Young's modulus are given in the table below.
A material with a large value of Y is relatively unstretchable; a large stress is required for a given strain.
For example, the value of Y for steel (2×10^{11} Pa) is much larger than that for rubber (5×10^{8} Pa).]

Substance	Density ρ ($kg\ m^{-3}$)	Young's modulus Y($10^9\ N\ m^{-2}$)	Ultimate Strength, S_u($10^6 Nm^{-2}$)	Yield strength, S_y($10^6 Nm^{-2}$)
Aluminium	2710	70	110	50
Copper	8890	110	400	200
Iron(wrought)	7800-7900	190	330	170
Steel	7860	200	400	250
Glass	2190	65	50	–
Concrete	2320	30	40	–
Wood	525	13	50	–
Bone	1900	9	170	–
Polystyrene	1050	3	48	–

Q. **A steel rod whose length is 2.0 m has area of cross-section 0.30 cm². The rod is now hung by one end from a support structure, and a generator of mass 550 kg is hung from the rod's lower end. Find the stress, strain, and the elongation of the rod.**

Sol. $\text{Stress} = \frac{F}{A} = \frac{(550\text{kg})\left(9.8\text{m/s}^2\right)}{3.0\times10^{-5}\,\text{m}^2}$

$= 1.8 \times 10^8$ Pa

$\text{Strain} = \frac{\Delta l}{l_0} = \frac{\text{Stress}}{Y} = \frac{1.8\times10^8\,\text{Pa}}{20\times10^{10}\,\text{Pa}}$

$= 9.0 \times 10^{-4}$

Elongation = Δl

$= (\text{strain}) \times l_0$

$= (9.0 \times 10^{-4})\ (2.0\ \text{m})$

$= 1.8$ mm

Q. **A structural steel rod has a radius of 10 mm and a length of 1.0 m. A 100 kN force stretches it along its length. Calculate (a) stress, (b) elongation and (c) strain on the rod. Young's modulus, of structural steel is 2.0×10^{11} Nm^{-2}.**

Sol. $\text{Stress} = \frac{F}{A} = \frac{F}{\pi r^2} = \frac{100\times10^3\,\text{N}}{3.14\times\left(10^{-2}\right)^2\,\text{m}^2} = 3.18\times10^8\,\text{Nm}^{-2}$

$\text{The elongation, } \Delta L = \frac{(F/A)L}{Y} = \frac{3.18\times10^8\,\text{Nm}^{-2}\times1\text{m}}{2\times10^{11}\,\text{Nm}^{-2}}$

$= 1.59 \times 10^{-3}$ m $= 1.59$ mm

$\text{Strain} = \frac{\Delta L}{L} = \frac{\left(1.59\times10^{-3}\,\text{m}\right)}{1\text{m}} = 1.59\times10^{-3} = 0.16\%$

POISSON'S RATIO

Any change in length is also accompanied with change in lateral dimension (breadth or radius of cross section). For a cylindrical rod, on applying a tensile force, if the increase in length is Δl, then radius decreases by ΔR.

So, the applied force is producing two strains in the rod.

(i) **Longitudinal strain** $= \frac{\Delta l}{l}$

(ii) **Lateral strain** $= -\frac{\Delta R}{R}$

Then, the Poisson's ratio, $\sigma = \frac{\text{Lateral strain}}{\text{Longitudinal strain}}$

$$= \frac{-\Delta R / R}{\Delta l / l}$$

$$= -\frac{\Delta R}{R} \times \frac{l}{\Delta l}$$

Here, negative sign shows that with increase in length, the radius of wire decreases.

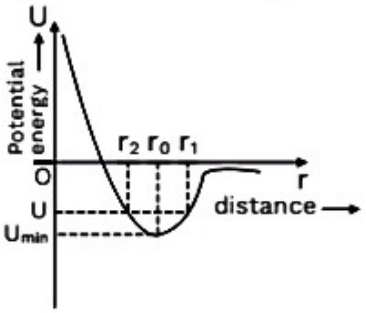
Poisson's ratio is a unitless and dimensionless quantity.

The theoretical value of Poisson's ratio for all substances lies in between -1 and +0.5, but practical value of Poisson's ratio lies between zero and +0.5.

Elastic Potential Energy

When elongating or compressing an object, the interatomic distance between atoms changes. It changes from equilibrium r_0 and either it increases to r_1 or decreases to r_2. .

Figure shows U-r graph between two atoms. The electrostatic potential energy between two atoms is minimum at $r=r_0$ which is stable equilibrium position.



For any other values of r, U will increase. It means that in a deformed body, the potential energy U between any two nearest atom or molecule is greater than the minimum value.

The difference in these two $|U-U_{min}|$ is the increase in energy between the atoms or molecules of a deformed body.
This extra interaction energy between all atoms/molecules of a deformed object is elastic energy.
On removing the external force, the body regains its original position and the elastic energy is retrieved back and converted into vibrations energy followed by heat, light, sound etc.

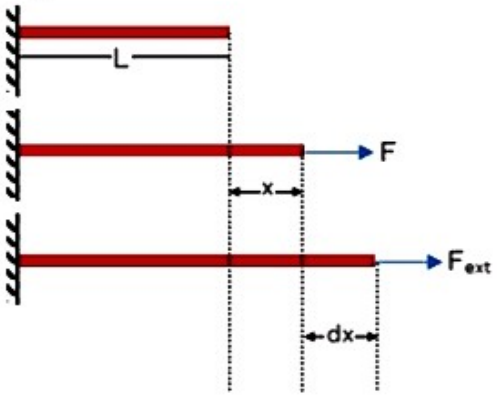
CALCULATION OF ELASTIC ENERGY

Consider a thin wire of Young's modulus Y, cross-sectional area A and length L. One of its ends is fixed and slowly pull the other end of the wire.

When the elongation of the wire is x, let the force required to pull the wire be F_{ext} .

Work done : While slowly elongating a thin wire, the external force has to be equal and opposite to the elastic force. Then

$$F_{ext} = F_{elastic} \quad \text{where } F_{elastic} = \frac{YA\cdot x}{L}$$



From definition of potential energy, $W_{ext} = -W_{elastic} = \Delta U$
Now, work done in stretching the rod by an elemental length dx against the restoring force will be, $dW = \frac{AY\cdot x}{L}dx$.

$\therefore$ Total work done against the restoring force in stretching the rod by a length l will be

$$W = \int_0^l dW = \frac{YA}{L}\int_0^l x\,dx = \frac{YA}{L}\left[\frac{x^2}{2}\right]_0^l$$

$$\Rightarrow \quad \boxed{W = \frac{YAl^2}{2L}}$$

$$\therefore \quad W = \frac{1}{2}\left(\frac{Yl}{L}\right)\left(\frac{l}{L}\right)(AL)$$

$$= \frac{1}{2} \times \text{Stress} \times \text{Strain} \times \text{Volume}$$

Thus, the elastic potential energy = W

$$\therefore \quad \Delta U = \frac{1}{2} \text{ stress} \times \text{strain} \times \text{volume}$$

And,

$$\boxed{\frac{\Delta U}{\text{Volume}} = \frac{1}{2}\left(\text{Stress} \times \text{Strain}\right)}$$

The above formula is valid for all types of strain.

Q. **A wire made of steel has a length of 4.0 m and a cross sectional area of 2.0 mm^2. It is elongated by 2.0 mm from its natural length. Using $Y_{Steel} = 2.0 \times 10^{11}$ Nm^{-2}, calculate,**

(i) Energy density of the wire.

(ii) Elastic potential energy stored in the wire.

Sol. Here, l = 4.0 m, $\Delta l = 2 \times 10^{-3}$m, $a = 2.0 \times 10^{-6}\ m^2$, $Y = 2.0 \times 10^{11}\ Nm^{-2}$

(i) The energy density of stretched wire,

$$\frac{\Delta U}{\text{Volume}} = \frac{1}{2} \text{ stress} \times \text{strain}$$

$$= \frac{1}{2} \times Y \times (\text{ strain })^2$$

$$= \frac{1}{2} \times 2.0 \times 10^{11} Nm^{-2} \times \left(\frac{2 \times 10^{-3}}{4}\right)^2$$

$$= 2.5 \times 10^4 J/m^3$$

(ii) Elastic potential energy = energy density × volume

$$= (2.5 \times 10^4 J/m^3) \times (8.0 \times 10^{-6}\ m^3)$$

$$= 0.20\ J$$

Q. **A area of cross section of bones of leg of a 45 kg boy is 5 cm² and the length of bone is 50 cm. The boy jumps from a height of 2 m without breaking his leg bones. If the bones can stand a stress of 0.9×10^8 N m^{-2}, calculate the Young's modulus for the material of the bone. Use g = 10 ms^{-2}.**

Sol. Loss in gravitational potential energy = Gain in elastic potential energy in both leg bones

$$\text{So, } mgh = 2 \times \left(\frac{1}{2} \text{ stress } \times \text{ strain } \times \text{ volume } \right)$$

$$(45 \times 10 \times 2) = 2 \times \left[\frac{1}{2} \times 0.9 \times 10^8 \times \text{Strain} \times 2.5 \times 10^{-4} \right]$$

Strain = 0.04

$$\therefore \quad Y = \frac{\text{Stress}}{\text{Strain}} = \frac{0.9 \times 10^8}{0.04} = 2.25 \times 10^9 \text{Nm}^{-2}$$

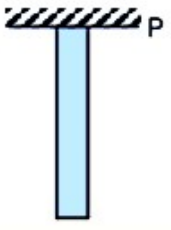
Q. **A metallic wire is stretched by suspending weight to it. If α is the longitudinal strain and Y is its Young's modulus of elasticity, show that the elastic potential energy per unit volume is given by $\frac{Y\alpha^2}{2}$.**

Sol. Stress = Young's modulus × Strain = Y × α
Elastic potential energy per unit volume is

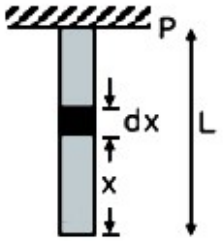
$$\frac{\Delta U}{\text{Volume}} = \frac{1}{2} \times \text{stress } \times \text{ strain } = \frac{1}{2} \times Y\alpha \times \alpha = \frac{1}{2} Y\alpha^2$$

Q. **A bar of mass M and length L is hanging from point P as shown in figure. The cross sectional area of wire is A and Y is its Young's modulus.**

(i) Find the stress at a distance x from the lower end.
(ii) Consider a small section dx of the bar at a distance x from lowest point of bar. Find elongation (dL) in section dx.
(iii) Find total elongation in bar.
(iv) Find energy density at distance x from bottom end.
(v) Find total elastic potential energy stored in bar..



Sol. (i) The weight of x length of the bar is



$$W = \left(\frac{Mg}{L}\right)x$$

So, stress at x distance from bottom,

$$\frac{W}{A} = \frac{Mgx}{AL}$$

(ii) Stress = Y (strain)

$$\frac{Mgx}{AL} = Y\frac{dL}{dx}$$

$$\Rightarrow \quad dL = \frac{Mgxdx}{ALY}$$

(iii) Total elongation in wire

$$\Delta L = \int_0^L dL$$

$$\Rightarrow \Delta L = \frac{Mg}{ALY}\int_0^L xdx$$

$$= \frac{MgL}{2AY}$$

(iv) Energy density at x distance,

$$= \frac{1}{2}(\text{ stress })(\text{ strain })$$

$$= \frac{1}{2}\frac{(\text{ stress })^2}{Y}$$

$$= \frac{1}{2Y}\left[\frac{Mgx}{AL}\right]^2 = \frac{M^2g^2x^2}{2YA^2L^2}$$

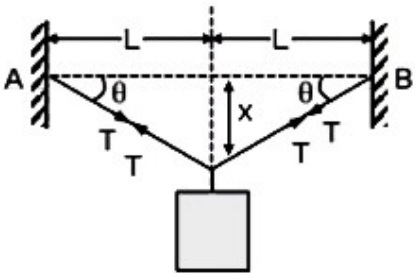
(v) Energy stored in Adx volume = Energy density × Adx = $\frac{M^2g^2x^2}{2YA^2L^2}(Adx)$

$$\text{Total energy } = \int_0^L \frac{M^2g^2x^2}{2YAL^2}dx = \frac{M^2g^2L}{6AY}$$

Q. **Consider a copper wire of radius r and length L is clamped firmly at two points A and B as shown. A body is suspended from the middle point of the wire such that middle point sags x meter lower from the original position. (If Young's modulus of material of wire is Y), then find out x.**

Sol. From figure,

$\sin\theta = \tan\theta = \frac{x}{L}$



By definition of Young's modulus,

$$T = \frac{YA\Delta L}{L} = \frac{YA}{L}\left[\left(L^2 + x^2\right)^{1/2} - L\right] = \frac{YAx^2}{2L^2}$$

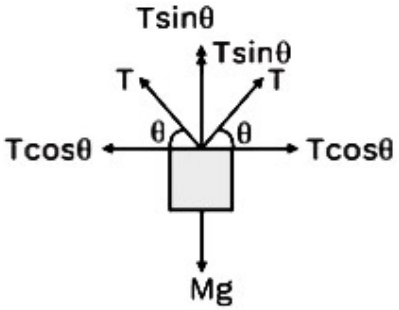
$2T\sin\theta = Mg$

Now, we can write that,

$$Mg = 2 \times \frac{YAx^2}{2L^2} \times \frac{x}{L}$$

$$Mg = \frac{YAx^3}{L^3}$$

$$\Rightarrow \boxed{x = \sqrt[3]{\left[\frac{M \cdot g \cdot L^3}{Y \cdot \pi r^2}\right]}}$$



Q. **A steel wire 6.0 m in length and 2.0 mm² area of cross section, is stretched through 3.0 mm. For $Y_{Steel} = 2.0 \times 10^{11}$ N/m², calculate-**
(a) The energy density of wire
(b) Elastic potential energy stored in the wire
(c) Find the minimum work done in stretching the wire

Sol. (a) The energy density of stretched wire,

$$U = \frac{1}{2} \times \text{stress} \times \text{strain}$$

$$= \frac{1}{2} \times Y \times (\text{strain})^2$$

$$= \frac{1}{2} \times 2.0 \times 10^{11} \left(\frac{3 \times 10^{-3}}{6}\right)^2$$

$$= 0.25 \times 10^5 = 2.5 \times 10^4 \text{J/m}^3$$

$$\Rightarrow \quad U = 2.5 \times 10^4 \text{ J/m}^3$$

(b) Elastic potential energy = energy density × volume

$= 2.5 \times 10^{4} \times (2.0 \times 10^{-6}) \times 6.0$

$= 20 \times 10^{-2} = 0.30\ J$

Elastic Potential energy = 0.30 J

(c) Work done = Potential Energy Stored

$$= \frac{1}{2}K(\Delta l)^2 \qquad \text{as } \left(K = \frac{YA}{l}\right)$$

$$= \frac{1}{2} \times \frac{YA}{l} \times (\Delta l)^2$$

$$= \frac{1}{2} \times \frac{2 \times 10^{11} \times 2 \times 10^{-6} \times 9 \times 10^{-6}}{6}$$

$$= 3 \times 10^{11} \times 10^{-12}$$

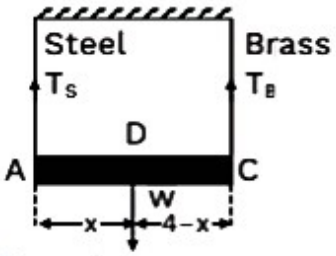
$$= 0.3\ J$$

Q. **A light rod of length 4.0 m is suspended from the ceiling horizontally by means of two vertical wires of equal length tied to its ends. One of the wires is made of steel and is of cross-section 10^{-3} m² and the other is of brass of cross-section 2×10^{-3} m². Find out the position along the rod at which a weight may be hung to produce equal stresses in both wires.**

Sol. Given, stress in steel = stress in brass

$$\frac{T_S}{A_S} = \frac{T_B}{A_B}$$

$$\frac{T_S}{T_B} = \frac{A_S}{A_B} = \frac{10^{-3}}{2 \times 10^{-3}} = \frac{1}{2}$$



As the system is in equilibrium, taking moments about D, we have

$$T_S \cdot x = T_B(4-x)$$

$$\frac{T_S}{T_B} = \frac{4-x}{x}$$

So, $$\frac{4-x}{x} = \frac{1}{2}$$

$$x = 8 - 2x$$

$$3x = 8$$

$$x = \frac{8}{3} = 2.67\text{m}$$

Position is 2.67 m from the end, where steel wire is attached.

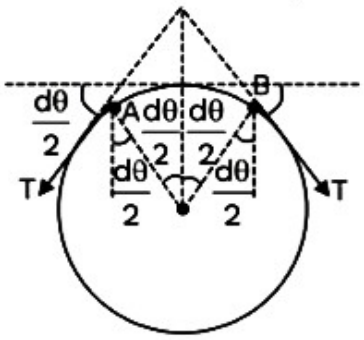
Q. **A thin ring of radius R is made of a material of density ρ and Young's modulus Y. If the ring is rotated about its centre in its own plane with angular velocity ω, find the increase in radius of the ring.**

Sol. onsider an element AB of length dl on the periphery of the ring.
Let T be the tension and A the area of cross section of the wire.
Mass of element, dm = Volume × density

$$= A(dl)\rho$$

The component of T, towards the centre provides the necessary centripetal fore

$$\therefore \quad 2T\sin\frac{d\theta}{2} = (dm)R\omega^2$$



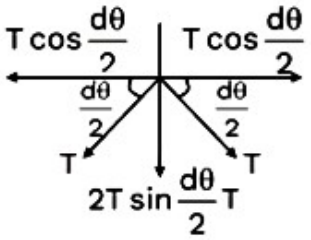
For small angle, $\sin\frac{d\theta}{2} \approx \frac{d\theta}{2} = \frac{1}{2}\left(\frac{dl}{R}\right)$

So, we can write,

$$2T\cdot\frac{d\theta}{2} = (dm)R\omega^2$$

$$T\cdot\frac{dl}{R} = A(dl)\rho R\omega^2$$

$$T = A\rho\omega^2R^2$$



Now, fractional change in radius,

$$\frac{\Delta R}{R} = \frac{T}{AY} = \frac{\rho\omega^2R^2}{Y} \quad \Rightarrow \Delta R = \frac{\rho\omega^2R^3}{Y}$$

Q. **A thin uniform rod of length L and mass m rotates uniformly with an angular velocity ω in a horizontal plane about a vertical axis passing through one of its ends. Find the elongation in the rod.**

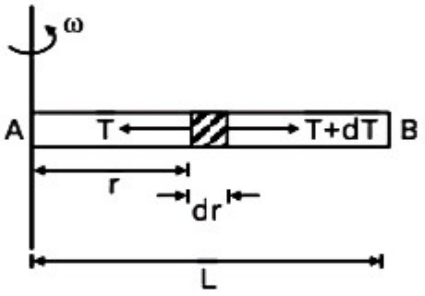
Sol. Every elemental mass on the rod moves in a circular path.
So, for circular motion to be possible, centripetal force in required which is provided by excess tension.
For the small element of length dr

$$T - (T + dT) = dm.\omega^2.r$$

$$-dT = \frac{m}{L}.dr.\omega^2.r$$

$$\int_T^0 -dT = \int_r^L \frac{m}{L}\omega^2 r\,dr \qquad (\text{At free end B, } T = 0)$$

$$T = \frac{m\omega^2}{L}\left(\frac{L^2 - r^2}{2}\right) \qquad T = \frac{m\omega^2L^2}{2L}\left(1 - \frac{r^2}{L^2}\right)$$



$$\boxed{T = \frac{m\omega^2 L}{2}\left(1 - \frac{r^2}{L^2}\right)} \qquad \text{....(i)}$$

Equation (i) gives tension (T) at a distance r from rotational axis.

If dl is elongation in the small element (of original length dr)

then, strain in this element is given by,

$$\text{Strain} = \frac{dl}{dr} \qquad \text{..(a)}$$

Also, $\text{Strain} = \frac{\text{Stress}}{Y} = \frac{T}{AY}$ (if A = Area of cross-section of rod)

$$\text{Strain} = \frac{m\omega^2 L}{2AY}\left(1 - \frac{r^2}{L^2}\right) \qquad \text{...(b)}$$

From (a) and (b),

$$\frac{dl}{dr} = \frac{m\omega^2 L}{2AY}\left(1 - \frac{r^2}{L^2}\right)$$

$$\int_0^{\Delta L} dl = \int_0^L \frac{m\cdot\omega^2\cdot L}{2AY}\left(1 - \frac{r^2}{L^2}\right)\cdot dr$$

$$\Delta L = \frac{m\omega^2 L}{2AY}\left(r - \frac{r^3}{3L^2}\right)_0^L = \frac{m\omega^2 L}{2AY}\left(L - \left(\frac{L^3}{3L^2}\right)\right)$$

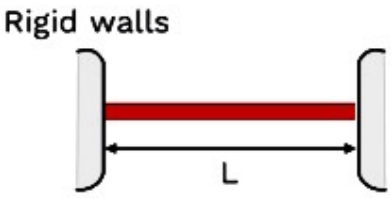
$$\Delta L = \frac{m\omega^2 L}{2AY}\left(L - \frac{L}{3}\right)$$

$$\Rightarrow \boxed{\Delta L = \frac{m\omega^2 L^2}{3AY}}$$

THERMAL STRESSES

A rod of length L is fixed between two rigid walls and is heated to raise its temperature by $\Delta\theta$. This causes an increase in its length (Δl).

But, the rod is fixed between rigid walls. So, it develops a compressive strain $\frac{\Delta L}{L}$.



Now, $\Delta l \propto L$

and $\Delta l \propto \Delta\theta$ (Change in temperature)

$\Rightarrow \Delta l \propto L\,\Delta\theta$

$\Rightarrow \Delta l = \alpha L\,\Delta\theta$ [where α is constant for a material]

$\Rightarrow \Delta l / L = \alpha\,\Delta\theta$ [Compressive strain]

As Y = stress / strain

$\Rightarrow$ Stress = Y (strain)

$\Rightarrow$ Stress = $Y\,\alpha\,\Delta\theta$, where α is the coefficient of linear expansion.

Q. **A steel rod of length 6.0 m and diameter 20 mm is fixed between two rigid supports. Determine the stress in the rod, when the temperature is increased by 80°C if ends do not break.**
Take $Y = 2.0 \times 10^6$ kg/cm² and $\alpha = 12 \times 10^{-6}$ per °C..

Sol. Let S be the stress in the rod.

Using the relation,

$$S = \alpha Y \Delta T$$

$$S = 12 \times 10^{-6} \times 2 \times 10^6 \times 80$$

$$= 1920 \text{ kg/cm}^2$$

Stress in the rod = 1920 kg/cm²

SHEAR MODULUS

If the forces are small enough that Hooke's law is obeyed, the shear strain is proportional to the shear stress.

The corresponding elastic modulus (ratio of shear stress to shear strain) is called the shear modulus, denoted by G. It is also called the modulus of rigidity.

$$G = \frac{\text{Shear stress}}{\text{Shear strain}} = \frac{\left(\frac{F_{||}}{A}\right)}{\left(\frac{x}{h}\right)}$$

$$G = \frac{F h}{A x}$$

Similarly, from equation,

$$G = \frac{F / A}{\theta}$$

The shear stress (σ_s) can also be expressed as

$$\sigma_s = G.\theta$$

SI unit of shear modulus in Nm^{-2} or Pa.

Q. **A 4 cm cube has its upper face displaced by 0.1 mm by a tangential force of 8 kN. Calculate the shear modulus of the cube.**

Sol. Here, side length of the cube, L = 4 cm

Area of the face over which the force is applied, $a = L^2 = 16\ cm^2$.

Displacement, $\Delta L = 0.1\ mm = 0.01\ cm$

Force applied, $F = 8\ kN = 8000\ N$

Now, $G = \frac{FL}{a\Delta L}$

$$G = \frac{8 \times 10^3 \times 4 \times 10^{-2}}{16 \times 10^{-4} \times 0.1 \times 10^{-3}} = 2 \times 10^9\ N/m^2$$

BULK MODULUS

When Hooke's law is obeyed, an increase in pressure (Bulk stress) produces a proportional Bulk strain (fractional change in volume).

The corresponding elastic modulus (ratio of stress to strain) is called the Bulk modulus, denoted by B.

When the pressure on a body changes by a small amount Δp, from p_0 to $p_0 + \Delta p$, and the resulting Bulk strain is $\Delta V/V_0$, then as per Hooke's law,

$$B = \frac{\text{Normal stress}}{\text{Volume strain}} = -\frac{\Delta P}{\Delta V / V_0} \quad \text{(Bulk modulus)}$$

We include a minus sing in this equation because an increase of pressure always causes a decrease in volume. In other words, if Δp is positive, ΔV is negative. The Bulk modulus B itself is a positive quantity.

The reciprocal of the Bulk modulus is called the compressibility and is denoted by K. From equation

$$K = \frac{1}{B} = \frac{-\Delta V / V_0}{\Delta p} = -\frac{1}{V_0}\frac{\Delta V}{\Delta p}$$

Compressibility is the fractional decrease in volume, $-\frac{\Delta V}{V_0}$ per unit increases Δp in pressure.

The units of compressibility are those of reciprocal pressure, Pa^{-1} or atm^{-1}.

Q. **A hydraulic press contains 0.25 m^3 (250 L) of oil. Find the decrease in the volume of the oil when it is subjected to a pressure increases $\Delta p = 1.6 \times 10^7$ Pa. The bulk modulus of the oil is $B = 5.0 \times 10^9$ Pa and its compressibility is $K = \frac{1}{B} = 20 \times 10^{-6}\ Pa^{-1}$.**

Sol. $$\Delta V = \frac{-V_0 \Delta p}{B} = \frac{0.25 m^3 \times 1.6 \times 10^7 Pa}{5.0 \times 10^9 Pa}$$

$$\Delta V = -8.0 \times 10^{-4} m^3 = -0.80 L$$

Q. **The average depth of Indian Ocean is about 3000 m. Calculate the fractional compression, $\frac{\Delta V}{V}$, of water at the bottom of the ocean, given that the bulk modulus of water is $2.2 \times 10^9 Nm^{-2}$. (take g = 10 m/s^2).**

Sol. The pressure exerted by a 3000 m column of water on the bottom layer

$P = \rho gh = 3000\ m \times 1000\ kgm^{-3} \times 10\ ms^{-2}$

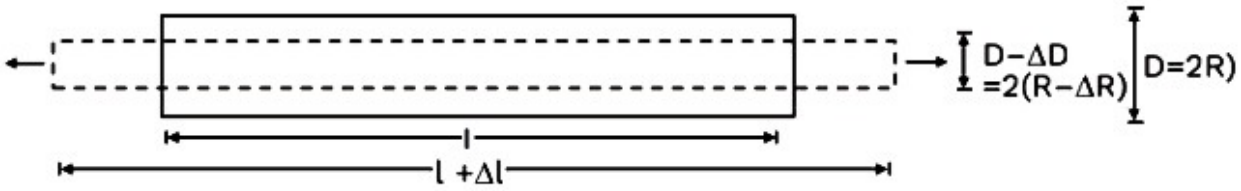
$= 3 \times 10^7\ Nm^{-2}$

Fractional compression $\frac{\Delta V}{V}$, is

$$\frac{\Delta V}{V} = \frac{\text{Stress}}{B} = \frac{\left(3 \times 10^7 Nm^{-2}\right)}{\left(2.2 \times 10^9 Nm^{-2}\right)} = 1.36 \times 10^{-2} \text{ or } 1.36\%$$

POISSON'S RATIO

When a body is linearly extended, it contracts in the direction at right angles. Poisson's ratio, σ is the ratio of the lateral strain to the longitudinal strain.



Longitudinal strain = $\Delta l/l$

Lateral strain = $-\Delta R/R$

The Poisson's ratio is given as,

$$\sigma = \text{lateral strain / longitudinal strain} = \frac{-\left(\frac{\Delta R}{R}\right)}{\left(\frac{\Delta l}{l}\right)}$$

−ve sign shows that if the length increases, then the radius of wire decreases.

Poisson's ratio is a unitless and dimensionless quantity.

The theoretical value of Poisson's ratio for all substance lies in between −1 and +0.5.

The practical value of Poisson's ratio lies between zero and 0.5.

Relation between Y, K, G and σ

1. $Y = 3K(1 - 2\sigma)$
2. $Y = 2G(1 + \sigma)$
3. $\sigma = (3K - 2G) / (2G + 6K)$
4. $9/Y = 1/K + 3/G$

APPLICATIONS OF ELASTIC BEHAVIOUR OF MATERIALS

In our daily life, most of the materials which we use, undergo some kind of stress. That is why, while designing a structure of the material we give due consideration to the possible stresses, the material might suffer at one stage or the other. The following examples illustrate this concept.

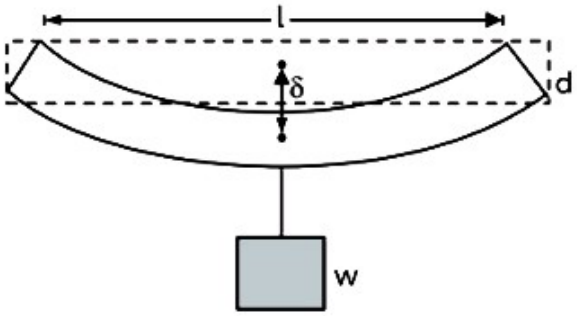
(1) The metallic parts of the machinery are never subjected to a stress beyond elastic limit, otherwise they will get permanently deformed.

(2) The crane which is used to lift and move the heavy load is provided with thick and strong metallic ropes to which the load to the lifted is attached. The rope is pulled by using pulley and motor. The thickness of the metallic rope used in the crane in order to lift a given load is decided from the knowledge of elastic limit of the material of the rope and the factor of safety.

(3) The bridges are designed in such a way that they do not bend much or break under the load of heavy traffic, force of strongly blowing wind and its own weight.

Consider a bar of length l, breadth b and depth d supported horizontally at its two ends as shown in figure. Let Y be the Young's modulus of material of the bar. When a load W is attached at its middle point, the depression δ produced at middle point in the bar is given by

$$\delta = \frac{Wl^3}{4Ybd^3}$$



(4) Maximum height of a mountain on earth ca be estimated from the elastic behavior of earth. At the base of mountain, the pressure is given by p =hρg, where h is the height of mountain, ρ is the density of material of mountain (= 3 × 10^3kg/m^3) and g is the acceleration due to gravity (= 10.0 ms^{-2}). At the base of mountain, the pressure must be less than elastic limit of earth's supporting material whose value for a typing rock is 3 × 10^8 Nm^{-2}. Therefore, hρg < 3 × 10^8 or h < $\frac{3 \times 10^8}{\rho g}$ or h < 10^4m, which is nearly, the height of Mount Everest.

Q. **What is the density of ocean water at a depth, where the pressure is 0.80 atm, given that its density at the surface is 1.03×10^3 kg/m^3 ? Compressibility of water = 45.8×10^{-11} Pa^{-1}. 1 atm = 1.013×10^5 Pa.**

Sol. Let ρ' is the density of water at a given depth.

Let for a given mass M, volume of water are V and V' at the surface and at the depth respectively.

Then,

$-\Delta V = V - V'$

$= M (1/\rho - 1/\rho')$

Volumetric strain

$-\Delta V/V = M (1/\rho - 1/\rho')/M/\rho$

$= 1 - \rho/\rho'$

$= 1 - 10.3 \times 10^3/\rho'$ ...(i)

Now bulk modulus

$K = -\rho V/\Delta V$

$\therefore \quad -\Delta V/V = p/K$

$= (80.0 \times 1.013 \times 10^5) \times 45.8 \times 10^{-11} = 3.712 \times 10^{-3}$ ...(ii)

Equating (i) and (ii) we have

$\rho' = 1.034 \times 10^3$ kg/m^3.